American Computer Science League

2024-2025 ● Contest 3: Short Problems ● Senior Division

1. Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$A\,(\overline{B} + \overline{C}) \;+\; \overline{A\,\overline{B}\,C} \;+\; \overline{B\,C}$$

A. 0
B. 2
C. 4
D. 6
E. 8

2. Boolean Algebra

Simplify the following Boolean expression:

$$\overline{A(B + \overline{C})}\;(A\,\overline{B\,C})\;(\overline{A} + \overline{B})$$

A. $\overline{A} * \overline{B} * C$
B. $A * B * C$
C. $A * \overline{B} * C$
D. $\overline{A} * \overline{B} * \overline{C}$
E. $A * \overline{B} * \overline{C}$

3. Data Structures

Build a binary search tree for:

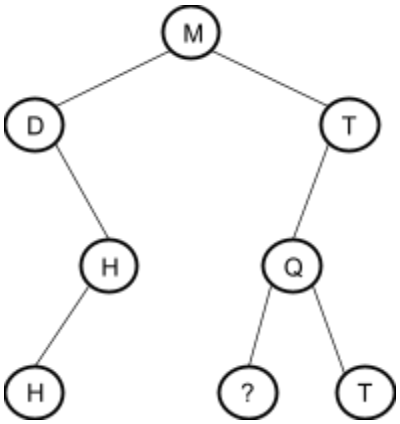
BEACHVOLLEYBALL

How many nodes have only one child?

A. 2
B. 4
C. 6
D. 8
E. 10

4. Data Structures

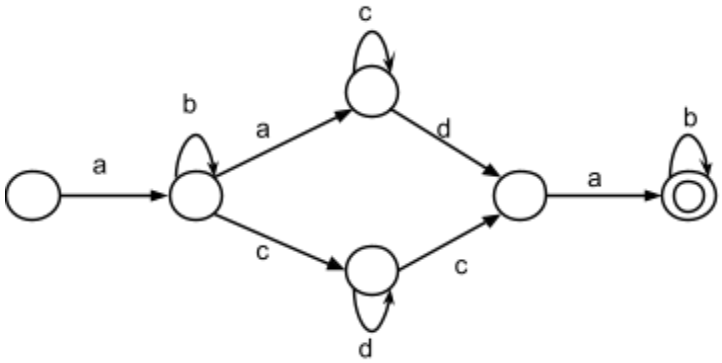
Given the following tree, find the values that can be used in place of the ?.



A. M < ? ≤ Q
B. M ≤ ? < Q
C. H < ? ≤ Q
D. M < ? ≤ T
E. D < ? ≤ T

5. FSAs

List all of the strings that are accepted by the following FSA:



a. abbada

b. abaccabb

c. acddcab

d. acca

A. b, c, d
B. a, b, d
C. a, b, c
D. a, c, d
E. a, b, c, d

6. Regular Expressions

The ACSL Board & Card Game Store opened this year. The games available must satisfy the following regular expression. How many will be available?

[a-cg-p]* [^s]* [^o-t] [d-ho-y]*

checkers	chess	monopoly	cribbage
bingo	uno	racko	duo

A. all of them
B. 7
C. half of them
D. 1
E. none of them

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1. 布尔代数

请问有多少个有序三元组可以使得下列布尔表达式的结果为真？

$$A\,(\overline{B}+\overline{C})\;+\;\overline{A\,\overline{B}\,C}\;+\;\overline{B\,C}$$

A. 0
B. 2
C. 4
D. 6
E. 8

2. 布尔代数

请简化下列布尔表达式：

$$\overline{A(B\;+\overline{C})}\;(A\,\overline{B\,C})\;(\overline{A}\;+\;\overline{B})$$

A. $\overline{A} * \overline{B} * C$
B. $A * B * C$
C. $A * \overline{B} * C$
D. $\overline{A} * \overline{B} * \overline{C}$
E. $A * \overline{B} * \overline{C}$

3. 数据结构

在以下字符串构成的二叉搜索树中：

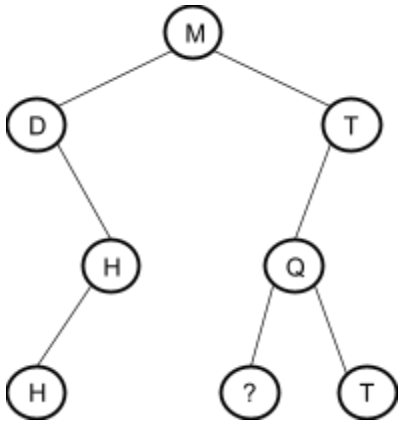
BEACHVOLLEYBALL

有多少个节点只有一个子节点？

A. 2
B. 4
C. 6
D. 8
E. 10

4. 数据结构

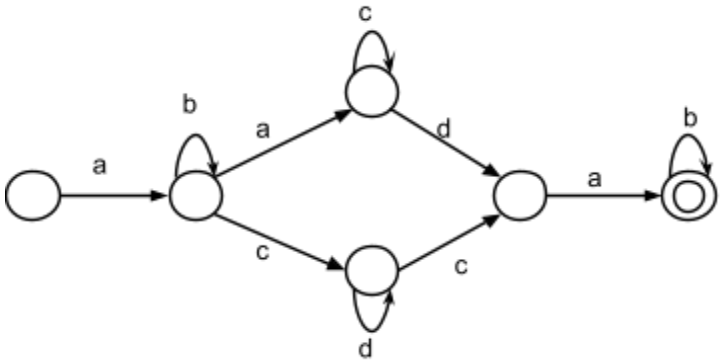
给定下列树形结构，**请**找出可以**填**在问号“ ？ ”处的值。



A. M < ? ≤ Q
B. M ≤ ? < Q
C. H < ? ≤ Q
D. M < ? ≤ T
E. D < ? ≤ T

5. FSAs

请问哪些字符串可以被**下列** FSA 接受？



a. abbada

b. abaccabb

c. acddcab

d. acca

A. b, c, d
B. a, b, d
C. a, b, c
D. a, c, d
E. a, b, c, d

6. 正则表达式

ACSL 桌游店今年开业。可供选择的游戏必须满足**下列**正则表达式。**请问**将会有多少款游戏可供选择？

[a-cg-p]* [^s]* [^o-t] [d-ho-y]*

checkers	chess	monopoly	cribbage
bingo	uno	racko	duo

A. 所有游戏可供选择
B. 7
C. 半数游戏可供选择
D. 1
E. 没有游戏可供选择